BrightLearn Data analytics
Research Assignment 1
Section A. Data Fundamentals

1. A database is an organized, digital collection of structured information or data, typically stored and accessed electronically from a computer system.
• A database organizes it in a way that makes searching and using the information quick and efficient.
• Databases are essential in the ~~morde~~ modern world because they help businesses and organizations store large amount of data safely and retrieve it quickly when needed, They also improve accuracy, save time, and allow multiple users to access the same information at the same time.
Examples.
* Banks use databases to store customer information, account balances, and transaction records.
* Schools use database to keep student records such as marks, ~~attrd~~ attendance and personal details.

2. A data warehouse is a large storage system that collects and stores data from different sources over a long period of time. it is mainly used for analyzing data and making business decisions, rather than for daily operations.
• A regular database is used for everyday tasks, such as entering, updating, and retrieving current data (for example, processing sales or student records). Databases focus on fast updates, while data warehouse focus on understading trends and patterns.
• It is using the data warehouse to, analyze large

amount of data from different departments.

- To identify trends business decisions.
- Make better business decisions.
- Improve planning and forecasting.

3. A Data ~~late~~ lake is a Storage System that keeps large amount of raw data in its original form. This data can be structured, Semi-Structured (like JSON files), or unstructured. The dat is stored as it is and can be processed later when needed.

- A data warehouse stores organized, structured data that has already been cleaned and prepared for analysis. Data warehouses are used mainly for reporting and business analysis, while data lakes are more flexible and can handle different type of data for deeper analysis.
- A business would choose a data lake when it need to store large volumes of different types of data.
- The data is not yet organized or structured.
- It want flexibility to analyze the data in different ways later.
- It is working with advanced analytics like machine learning.

4. A data mart is a smaller, focused section of a data warehouse that is designed for a specific department or group within an organization. It contains only the data that is relevant to that particular area, making it easier and faster to use.

- A data mart is usually created from a larger data warehouse. a data warehouse is the main storage, and data marts are smaller parts of it, while the data warehouse stores all the company's data in one place, the data mart

provides a smaller, more specific view of that data for a particular purpose.

- Data mart is used by specific department such as, Sales teams, Marketing teams and finance teams.

5. Structured Data

- Structured data is organized in a clear format, usually in rows and columns (like a table). It is easy to store, search and analyze because it follows a fixed structure. Example: A spreadsheet of student marks where each row represents a student and each column shows subjects and scores.

* Semi - Structured data

- Semi - Structured data is not stored in a strict table format, but it still has some level of organization using tags or labels. It is more flexible than structured data. Example: Emails, where you have fields like sender, receiver, subject, and message body.

* Unstructured data

- Unstructured Data has no specific format or structure. It is harder to organize and analyze because it does not follow a predefined model. Example Videos, images, or social media posts.

6. Data modeling is the process of creating a visual or logical representation of how data is structured, stored and related within a system.

- It important because, helps organize data clearly before building a stystem, Reduces errors and duplication of data.

- Types of Data models: Conceptual data model - This is a high-level model that shows the main entities like people, Places or things and the relationships between them.

Logical Data Model - This model is more detailed and shows attributes (data field) and relationships but it is not tied to a specific database system.

7. Data mining is the process of analyzing large sets of data to discover hidden patterns, and useful information.

- It is used in business by understand customer behavior improve decision making and predict future trends.

* Retail stores Supermarkets analyze customer purchase data to find patterns, This helps them arrange products in store and create special offers.

* Banks. They use data mining to detect unusual transactions and prevent fraud by identifying patterns that do not match normal customer behavior.

8. A Database Management System (DBMS) is software that allows users to create, store, manage, and retrieve data in a database.

* MySQL - is a widely used relational DBMS that stores data in tables with rows and columns.

* PostgreSQL - is an advanced relational DBMS that supports complex queries and large amount of data.

* Oracle Database - is a powerful, enterprise-level DBMS used by large organizations.

* MongoDB - is a NoSQL DBMS that stores data in flexible, JSON-like documents instead of tables, it is good for handling unstructured or rapidly changing data.

1. ETL = Extract, Transform, Load

Extract → get data

Transform → Clean/Format

- Load → store

10. Data Formats

- CSV → simple tables
- JSON → APIs
- Parquet → big data
- Avro → Streaming

Section B: AI & Machine Learning

11. Artificial Intelligence

- AI is the ability of machines to perform tasks that require human intelligence.
- Narrow AI: specific tasks (e.g. chatbots)
- General AI: human-level intelligence (still theoretical)

12. Machine Learning

- ML allows machines to learn from data.

Difference:

- Traditional → rules programmed
- ML → Learns pattern

Types:
* Supervised
* Unsupervised
* Reinforcement

13. Deep Learning

- Deep Learning is a type of ML using neural networks.
- Neural Networks: systems inspired by the human brain that process data in layers.

14. NLP - Natural Language Processing allows machines to understand human language.

Example:
- voice assistants
- chatbots.
- Translation apps.

15. LLM - Large Language Models are advanced AI models trained on large text data.
Examples. GPT chatGPT → Conversation
. Claude → Writing and analysis

16. General AI - It is creates New content
Tools. ChatGPT → text
. DALL-E → Images
. Midjourney → art

17. Training Dataset - A dataset used to train AI
Importance. Bad data = biased results.

18. Computer Vision - Allows machines to Interpret Images
Examples. face recognition
. Self - driving cars

Section C . APIs & TOOLS

19. API - An API allows different software systems to communicate.
Analogy. like Waiter taking orders between you and the kitchen.

20. API key - A unique code used to access an API
Security. Must be kept Private.

21. Rest API - A unique code used to access an API.

21. Rest API -
= REST - Representational State Transfer.

Methods : Get → retrieve
. Post → create
. Put → update
. Delete → remove

22. JSON - A format for storing and exchanging data
Example -
{ "name" : "John",
"age" : 23
}

23. Prompt Engineering
Bad : "Tell me about AI".
Good : "Explain AI in simple terms with ~~exapt~~ examples for beginners".

24. MAP function - Applies a function to each item in a list.
Example (Python). numbers = (1,2,3)
result = list (map (lambda x : x*2, numbers))

25. Cloud computing - Delivery of computing services over the internet.
Modules. Iaas → AWS
. Paas → Google App Engine
. Saas → Gmail

26. Version control - Track changes in code.
Git. popular system
• Commit → Save changes
• Branch → Separate versions

27. Jupter Notebook - A tool for writing and running code.
Language : Python
Why Popular : Easy for analysis and visualization.

28. Python - A programming used in data analytics.
Libraries : Pandas → data analysis
. NumPy → Math
. MatPlotlib → graphs
. Scikit learn → ML

Section D : DATA ETHICS

29. Data Privacy - Protecting personal data.
GDPR : A Law in Europe protecting user data.

30. Data Ethics
Examples : Selling personal data
. Based AI decisions
. Surveillance misuse

31. Data Governance - Managing data quality, Security, and usage.
Components : Policies
. Standards
. Security

32. Bias in AI - When AI produces unfair results.
Example : Hiring System favoring certain groups.

33. Roles Comparison

Role	Focus	Skills
Data analyst	Analyze data	Excel, SQL
Data Scientist	Predict models	Python, ML

Role	Focus	Skills
Data Engineer	Build Systems	Databases
AI Engineer	AI Systems	ML, DL

34. Business Intelligence - BI helps analyze data for decisions.
Tools . Power BI
. tableau

Section E . Personal Reflection (sample)

35. Most Interesting
• AI and machine learning because they are shaping the future.

36. Most Difficult
• Deep Learning because it involves complex concepts.
Resources used . YouTube, Google, Course notes.

3.7. Future Impact
Data and AI will help in careers by improving decision-making and automation.
Examples using AI tools for coding and analysis.

References

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- AWS. (2023). Cloud Computing
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number 9: ETL Process:

Extract	Transform	Load
Collect data	Processing data	storing data
Databases	Data cleaning	Data warehouse
Web/APIs	Data Intergration	Analytics
flat files	Data Transformation	BI Tools/cloud